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Implementation of QKD BB84 Protocol in Qiskit

Muhammad Haroon Saeed
National Institute of Lasers and
Optronics College, Pakistan
Institute of Engineering and
Applied Science, Nilore,
Islamabad 45650, Pakistan
haroonsaeed262@gmail.com

Hassan Sattar
Department of Computer &
Information Sciences, Pakistan
Institute of Engineering and
Applied Sciences, Nilore,
Islamabad 45650, Pakistan
bscs1724@pieas.edu.pk

Muhammad Hanif Durad
Department of Computer &
Information Sciences, Pakistan
Institute of Engineering and
Applied Sciences, Nilore,
Islamabad 45650, Pakistan
hanif@pieas.edu.pk

Zeshan Haider
National Institute of Lasers and
Optronics College, Pakistan
Institute of Engineering and
Applied Science, Nilore,
Islamabad 45650, Pakistan
shani12441@gmail.com

Abstract— Quantum cryptography is a practical method of secret communication that guarantees the foolproof secrecy. This paper presents the simulation for implementation of a Quantum Key Distribution (QKD) protocol BB84, the pioneering, and most basic and fundamental protocol. We use the quantum computing platform developed by IBM named as Qiskit for the simulation of the BB84 protocol, implementing it with and without an Eve eavesdropper for studying the relative outcomes. Recently, Qiskit is globally very popular for quantum simulations because it provide us the opportunity to simulate quantum logic circuit on classical computer as well as real quantum computers owned by IBM. As per our best knowledge, there is no any proposal published for quantum circuit by anyone for BB84 QKD. We performed experimentation on local as well as cloud simulators: 1) The “qasm_simulator” was used as local machine simulator, and 2) The “ibmqx2” simulator was used as IBM cloud quantum simulator. The outcome of local as well as IBM simulator with and without an eavesdropper, between the two parties involved in key exchange were recorded and presented in the results section.

Keywords— Quantum Key Distribution, Quantum Cryptography, Qubits , BB84 Protocol

I. INTRODUCTION

Traditional and classical cryptography has worked for quite a long time and is still broadly being used, however, with the advent of quantum technologies especially quantum computers, all these classical crypto-system are rendered insecure and hence unreliable [1], [2].

Now, in order to guarantee security and secrecy, one would need to utilize a totally irregular and truly random key, long as much as the plaintext, and never use it repeatedly, a code known as one-time Pad (OTP) [3]. When this arbitrary key is produced, one can apply for instance the bitwise XOR to the plaintext and, being the XOR symmetrical, recipient simply needs to reapply the XOR with a same key to the ciphertext to get the plaintext. OTP has, however, never acquired widespread popularity for one major practical reason: there is no 100% secure way to distribute the random key among the communicating parties [4]. It is precisely this point where quantum theory enters into cryptographic domain backed by the entirely counterintuitive and nonlocal traits of the superposition states [5]. One of the most widely use and pioneering protocol for key distribution using quantum mechanics is BB84 [6]. To be more exact, the

security and secrecy of BB84 protocol that utilizes polarization superposition states of the single photons for key distribution between the two parties relies fundamentally upon two basic features of the quantum theory i.e. (i) Quantum measurement always disturb a quantum state [7] (ii) Quantum states cannot be cloned or copied [8]. Basic physics of the BB84 protocol is well known and can be found in many good references [9], [10]. The present article, however discusses it very briefly, as can be seem in the next sections.

Our end results will be focused on compiling, executing BB84 key exchange protocol developed in Python on a Quantum Computer from IBM Quantum Experience and recording results for comparisons with simulator based results. We will be implementing BB84 in Python Qiskit [11], executing it in simulators and IBM cloud quantum service, recording the observed changes in outcomes in both platforms with and without an eavesdropper entity between the two parties involved in key exchange. We will be developing the Qubits sharing strategy for Alice, Bob and Eve entities involved in the key distribution.

Organization of this paper is as follow; Section II contains review of core concepts useful for understanding cryptography. In Section III, the necessary quantum properties explained briefly. Section IV is mainly on brief introduction of BB84 protocol of quantum key distribution (QKD). In Section V the important tools for quantum simulation presented. The results of our simulation of QKD BB84 protocol perform on Qiskit covers in the Section VI and the last Section is dedicated for summary and future work.

II. REVIEW OF CORE CONCEPTS

A. Symmetric and Asymmetric Cryptography

Encryption is an instrument to change the type of any content to keep it from being read by others. In Symmetric-key encryption (otherwise called secret-key cryptography), the message is then encrypted utilizing a key and with same key, the messages can be decrypted, simplifying it to utilize however less protected. It also needs a secure way of moving the key from one party to another [12]. BB84 protocol discussed here belong to this category but with the guaranteed secure solution to the key distribution problem.

The size of ciphertext is smaller or same as plaintext, with encryption measure being amazingly fast, transferring enormous amounts of information and having an extremely low resource utilization. Symmetric-key encryption can just give confidentiality. AES, DES, 3DES, RC4 all uses symmetric-key encryption [13,14].

Asymmetric Key Encryption (otherwise called public-key cryptography) is for both for public and private key encryption procedure. It utilizes separate keys to encrypt and decrypt a message or document [13]. It is more reliable than the symmetric key encryption strategy, however it is slower.

The size of ciphertext is larger or same than plaintext and appropriate for sending small amount of data only. Asymmetric key encryption gives both authenticity and confidentiality. ECC, El Gamal, Diffie-Hellman, DSA, RSA all uses Asymmetric-key encryption. Most of such classical protocols relies on the factorization intractability of large prime numbers [14]. However, quantum computers have recently shown their potential to solve such hitherto intractable problems [2].

B. Key Exchange

Key exchange is the process of sharing keys between two parties permitting them to encrypt information for communications [14]. Since we are dealing with ciphers rather than codes, the two parties will require suitable keys rather than a codebook. The nature of keys relies on the decision and properties of the cipher that is being utilized.

As referenced previously, a symmetric key cryptography requires the two parties to have duplicates of a same key while asymmetric key cryptography requires to have a Public key for the users and Private key for the party decrypting the coded messages [14].

Notable key exchange protocols are as follows:

RSA: As already mentioned, RSA is a public key cryptosystem that utilizes the practical difficulty of factoring product of two large prime numbers. It uses public encryption key meaning anyone can encrypt data but since it uses a private decryption key therefore only the one who knows the prime numbers can decrypt the data. [14].

Diffie-Hellman: is a method that enables securely sharing encryption keys on an insecure public channel. It is possible to do so by sharing the public keys of both parties with each other, each party combining their own private key with the public key of the other party to create a shared key that can be then used for symmetric encryption. [14].

C. Simulations

In usual terminology, simulator is a machine or piece of equipment designed to test cases before executing in the real world environments while simulation is the process of imitating a scenario [15]; Example could be a cockpit simulator of an aircraft that simulates weather conditions, drag forces, controls and haptic feedbacks to train or test pilots before being in real aircraft with passengers on board [15].

III. QUANTUM PROPERTIES

In Quantum computing, a qubit or a superposition state is the basic unit of quantum information. In classical computing, a bit can either be in state 0 or state 1 but never both simultaneously, however quantum mechanics allows the two states i.e. 0 and 1 to be exist at same time and this coherent superposition state is

known as qubit, a fundamental pre-requisite for quantum informatics [16].

Storing information in the form of qubits is the first step in implementing quantum cryptography, the science of safely exchanging information using quantum physics. This is achieved through the establishment of a quantum channel [14], a special apparatus capable of delivering quantum information in the form of qubits. Confidentiality is guaranteed by laws of quantum mechanics, a remarkable way of calling physics in aid of mathematics.

The basic quantum bits can be represented as $|0\rangle$ (pronounced as “Ket 0”) and $|1\rangle$ (pronounced as “Ket 1”) [16].

To represent the coherent superposition of the qubit, we can describe it by a linear combination of ket 0 and ket 1 as;

$$|\Psi\rangle = \alpha|0\rangle + \beta|1\rangle \quad (1)$$

Here the probability amplitudes are represented by α and β and yield probabilities when squared.

Light in nature contains all of its components, for obtaining a single component of the light we use polarizers that can be of 0° , 45° , 90° , 135° . We form two basis from pairs of these polarizers. A Diagonal Basis using polarizers in 45° and 135° , a Rectilinear Basis using polarizers in 0° and 90° [10,16].

A. Quantum Superposition

Quantum Superposition property describes that for two or three or any arbitrary quantum basis states can be superposed or added together and the resultant state will again be a lawful, valid quantum state e.g. qubit, qutrit and qudit etc. Alternatively, any valid quantum state can also be defined as sum of individual distinct states. In other words, a quantum particle can be in two or more states at the same moment of time [16]. In BB84, we employ polarization superposition state of a photon.

B. No Cloning Theorem

No-cloning or no-copying theorem simply asserts that faithful copying of an unknown quantum state is strictly forbidden according to the theory and hence we, in general, cannot clone or exactly copy any given quantum state [8,16]. The theorem blocks all the copying strategies after employed in classical cryptography.

In traditional electronics, digital signals are carried by conductive wires or traces in which current flows. If we wish to clone an arbitrary unknown signal on another track, the only thing we need to do is to bifurcate the wire and, given the right accuracy on the circuitry, we end up with the same signal duplicated on two tracks [14]. In quantum computers things work differently due to no-cloning theorem, something unforeseeable in the classical world. Thus, if the state is in an arbitrary state, we have no way of making exact copies of it. This is due to the fact that, when we work with quantum systems, we are broadly limited to performing only two fundamental operation on any quantum state. Firstly, we can strongly measure the quantum state and such a unitarity breaking operation consequently reduces the quantum state yielding as the data corresponding to the system’s eigen state only and hence the information contained or carried by the coherent nature of the state is lost or we can utilize a unitary operation [17].

Further, the no-cloning theorem has great implications in quantum computing. It prevents the usual classical methodology of correcting the incumbent errors in the quantum states. This is because one cannot retain the copies of a state going through unitary evolution in the midst of the quantum manipulation and processing. In classical computation, one always use such copies for error correction, a technique entirely unavailable in quantum domain on strict theoretical basis. It seems that quantum no-cloning theorem lies at the core of the quantum theory because the same theorem forbids superluminal communications and hence guarantees the peaceful co-existence of the relativity theory and the quantum mechanics, both being the pillars of the modern physics. Thus it is the no-cloning theorem that ensures no-signaling condition even in the presence of nonlocal entangled states. The most relevant consequence of the no-cloning theorem for our case is the fact that it's impossible to make copies or even to distinguish non-orthogonal states with only one measurement [16], [17].

C. Qubits

A quantum channel provides a medium and an apparatus to exchange quantum information between two parties using qubits, a set of two state quantum mechanical systems [16], [17]. There are many kinds of qubits such as electrons, photons, quantum dots, Josephson junctions etc. Polarized photons were the first type of qubits to be used in practice and they continue to be popular today on practical grounds, since they are relatively easy to obtain compared to other qubits, and they can be exchanged either through fiber optic cables or free space. In classical physics light is an electromagnetic waves, when quantized will becomes the stream of photons, quantum particle of light (or electromagnetic waves). The term polarization classically donates the directionality of wave's electric field [8], but can be also seen as an intrinsic property of single photons. Natural and commonly used sources of the electromagnetic radiations are often found in the form of unpolarized and incoherent light. This is because most of such sources generally mix up of diverse frequencies, phases and electric field vector orientations corresponding to a randomized set of polarization states. Now the conventionally employed strategy to get polarized light is either to use a polarization filter that blocks all undesired components and only allows a specific polarization to pass through or alternatively to utilize a laser with Brewster angle window generating coherent and polarized light. [17].

D. Quantum Gates

A quantum gate or quantum logic gate is a simple quantum circuit working on few qubits. They are the analogs for quantum computers to classical logic gates for regular digital computers. Quantum logic gates are reversible, not like many classical logic gates. Some universal classical logic gates, Toffoli gate for instance, gives reversibility and can be directly mapped onto quantum logic gates. Quantum logic gates are represented by unitary matrices [17].

The common quantum gates usually work on spaces of at most two qubits. Implying that quantum gates can be depicted in matrices as 2×2 or 4×4 with orthonormal rows.

Pauli-X gate: The Pauli-X gate is a single-qubit rotation through π radians around the x-axis.

Pauli-Y gate: The Pauli-Y gate is a single-qubit rotation through π radians around the y-axis.

Pauli-X (X)			$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$
Pauli-Y (Y)			$\begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$
Pauli-Z (Z)			$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$

Pauli-Z gate: The Pauli-Z gate is a single-qubit rotation through π radians around the z-axis.

Hadamard (H)		$\frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$
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The Hadamard gate is a single-qubit operation that maps the basis state $|0\rangle$ to $(|0\rangle + |1\rangle)/\sqrt{2}$ and $|1\rangle$ to $(|0\rangle - |1\rangle)/\sqrt{2}$, thus creating an equal superposition of the two basis states.

IV. BB84 QKD PROTOCOL

BB84 is a quantum key distribution scheme developed by Charles Bennett and Gilles Brassard in 1984 [6]. It is the pioneering quantum cryptography protocol. The protocol is provably secure, relying on the quantum property that information gain is only possible at the expense of disturbing the signal. It is usually explained as a method of securely communicating a private key from one party to another.

Although practical quantum computing is yet underdevelopment, the quantum cryptography is already a reality and products are out in the market pioneered by firms like ID Quantique, MagiQ Technologies, SmartQuantum and there are research groups working on it at corporations like HP, IBM, and Toshiba [18].

Security in software or online applications is generally provided by classical encryption algorithms like DiffieHellman, El-Gamal, RSA, and ECC [13], but all of these are either breakable or subject to cryptanalysis if sufficient technological resources were available. In quantum cryptography, on the other hand, it can be proved that Eve cannot access information without Alice (sender) and Bob's (recipient) knowledge.

BB84 and its variants are now most popular QKD protocols and use quantum bits in one pass and this is followed by two additional passes of classical data transmission.

A. BB84 Protocol (without Evesdropper)

We'll be dividing the process of sharing and creating the key between Alice and Bob in to six total steps.

1. Alice generates a random-string which is a key that will be shared with Bob.
2. Alice converts the string bits into its corresponding qubits. If bit is a 0, Alice prepares qubit on negative Z axis, otherwise on positive Z axis. If Alice has 1 in her rotate string, she rotates her key qubit with Hadamard gate.
3. Alice sends those qubits to Bob, randomly rotating some into superposition, effectively turning key into random noise.
4. Bob receives the qubits and randomly rotates some qubits in the opposite direction before measuring them. If Bob has

a 1 in his rotate string, he rotates the incoming qubit in opposite direction with Hadamard gate. Bob then measures the state of the qubit and records the result.

- 5 Alice and Bob will publicly share which qubits were rotated, they will know the original key bit value made it to Bob when both have used the same transformations or same basis that is either both did nothing or both rotated. As a result, only some bits of the Alice's original key have made it.
- 6 Alice and Bob create their keys, both keeping the bits that they know made it to the other one.

B. BB84 Protocol (with Evesdropper)

We'll be dividing the process of sharing and creating the key between Alice and Bob in eight total steps, this time with an eavesdropper named Eve.

- 1 Alice generates a random-string for creating a key, like before she will convert the string bits to corresponding qubits.
- 2 Alice sends her qubit to Bob but Eve measures the results.
- 3 To avoid suspicion, Eve prepares qubits corresponding to the bits she measured and sends them to Bob. Since Eve doesn't know which of the qubits she measured were in superposition or not, she doesn't even know whether to send exact values she measured or opposite values. In the end, sending on exact values is just as good as a deception as mixing them up again.
- 4 Alice and Bob randomly select the same parts of their keys to share publicly.
- 5 If the selected part of the keys did not match, they know Eve was eavesdropping..
- 6 If the selected part of the keys did match, they can be confident that the Eave was not eavesdropping.
- 7 They throw away the part of the key they made public and keep encrypting and decrypting messages with the portion of the key they have left.

C. Quantum Key Distribution

Let us assume that an intruder or eavesdropper conventionally termed as Eve is trying to decode the message by intercepting the secret communication between the legal parties conventionally called Alice and Bob. This interception if carried over the quantum channel is doomed to fail because quantum key distribution is protected by two very fundamental principles of the quantum theory which are namely the no-cloning theorem and the measurement induced detectable changes in the quantum state [17]. Further, the quantum discrimination theory states that it is impossible to differentiate or discriminate two mutually non-orthogonal states and hence it is no way possible to acquire the desired, coded information without imploding at least the quantum state of the one qubit under consideration.

Let us see how no-cloning theorem forbids such a action by blocking the copying option. Assume that we are given two states $|\psi\rangle$ and $|\phi\rangle$, respectively. We further assume that these states are, in general, mutually non-orthogonal. Now, suppose that Eve is attempting to acquire the information encoded into such states by copying them over an initialized ancilla qubit $|\Sigma\rangle$. The cloning machine works unitarily as under;

$$|\psi\rangle \otimes |\Sigma\rangle \longrightarrow |\psi\rangle \otimes |\xi\rangle$$

$$|\phi\rangle \otimes |\Sigma\rangle \longrightarrow |\psi\rangle \otimes |\xi'\rangle$$

Now, Eve only gets useful information when $|\xi\rangle$ and $|\xi'\rangle$ are different and true copies of $|\psi\rangle$ and $|\phi\rangle$, respectively. Now it is well known that inner product is totally preserved for the unitarily evolving quantum systems. This implies that;

$$\langle \xi | \xi' \rangle \langle \psi | \phi \rangle = \langle \Sigma | \Sigma \rangle \langle \psi | \phi \rangle \quad (2)$$

$$\langle \xi | \xi' \rangle = \langle \Sigma | \Sigma \rangle = 1 \quad (3)$$

Hence, $|\xi\rangle$ and $|\xi'\rangle$ strictly needs to be identical and therefore one can easily conclude that Eve should always have to disturb irreparably at least one of the state in order to acquire the desired information.

V. IMPORTANT TOOLS FOR QUANTUM SIMULATION

A. IBM Quantum Experience

The Quantum composer developed by the IBM venture, earlier commonly termed as the IBM Quantum Experience offers a real-time, online accessibility publically to use IBM's cloud based quantum computing facility [19]. This free offer includes access to prototype quantum processing facility developed by the IBM along with the instructions, tutorials and interactive textbook for helping out the beginners. As of the early months of the year 2021, about 20 quantum computational gadgets were in operation out of which six were dedicated for free computational services available to the working public. Such a free service can be effectively utilized to solve and simulate computational scenarios as well as to get good tracing through the available tutorials and related resources on the emerging subject of the quantum computing.

Taking of the hardware, the IBM quantum machines incorporates qubits comprised of superconducting transmon. In order to avoid decoherence, a major threat to quantum informatics, along with other noises and errors, IBM has employed a dilution refrigerator that operates at extremely low temperatures. These Quantum computers are installed at IBM research facility situated at the Thomas J. Watson Research center. Here the users can easily interact with quantum computer through the circuit model of the computation employed by the IBM. Such circuits are structured either by using graphical interface linked with the quantum composer or alternatively utilizing programming option offered in the Python's Jupyter notebook available as a guideline for the quantum lab. Circuits are generally designed by using Qiskit tools and are compiled through OpenQASM prior to real execution over a true Quantum computing system.

This incorporates admittance to a bunch of IBM's prototype quantum processors, a bunch of tutorials on quantum computation, and access to an interactive textbook. As of February 2021, there are more than 20 devices on the service, six of which are freely accessible by public. This service can be utilized to run algorithms and experiments, and explore tutorials and simulations around what may be possible with quantum computing [19].

B. Python Qiskit

Qiskit is an openly accessible open source for utilizing quantum computers for data processing [11,20]. Qiskit permits simple and innovative as well as easy research conduction for industrial application that have the very high potential for

quantum benefits. It's a standard framework of very high multi-tasking quantum algorithms and have applications for short term quantum computing problems. Qiskit utilizes Python 3.6 or later versions. However, one must keep in mind that both Qiskit and Python are continuously under development independently and issues of compatibility may occur with time.

Python is an efficient and interpreted high-level generic programming language and its structural morphology pay high attention on code readability with its vitally important utilization of indentation [20].

The most suitable and widely utilized approach is to use Python's Jupyter notebook. In such a notebook, all the data, whether dynamic or static, is incorporated into two categories i.e. code cells and text cells.

VI. RESULTS AND ANALYSIS OF PRESENT WORK

The BB84 algorithm was executed on classical as well as IBM quantum simulator. The Qiskit library provides different types of classical simulators, we used the "qasm_simulator". The IBM-Q machine also supports variety of simulators, we used "ibmqx2". On both simulators, Alice represents the sender, Bob represents the receiver and Eve represents the intruder. The following two experiments were performed one by one on both simulators:

- 1) Without Eve: In this experiment, Alice sends message to Bob using quantum channels without any intruder.
- 2) With Eve: In this experiment, Alice sends message to Bob but Eve intercepts quantum channel as well.

A. Experiments using Classical Simulator

In first case we run our experiment on classical simulator. The actual quantum computers/simulators that are freely available access by a lot of researchers of the whole world at a time, therefore, they are not easily available. So, we do not send our code directly to quantum simulator for execution if there are any syntax or logical errors in the code. We instead use classical simulator for verification whether our source code is logically correct or not. Once our code shows promising results we then put our code in queue for execution on real quantum computer.

The source code is almost same for both cases. We use both the options i.e. quantum and classical simulators, through backends. Backends represent either a classical simulator or real quantum computer and are responsible for running quantum circuits and returning the results.

The brief detail for algorithm of the quantum circuit is as given in this paragraph. Alice sends a quantum circuit as message to Bob on which he performs the measurements after receiving and gets the information in terms of probability distribution of classical signals, as shown in Figure 1. For n qubits ($n = 4$ in our case), we initialize the n quantum and classical registers associated with the quantum circuits of the Alice and the Bob. Then, we generate the random numbers, we are going to use them as a key. Alice assign the random bit values to her qubits using generated random numbers. In next step, Alice randomly apply Hadamard-gate on her 50% of qubits to transform them into the rotated basis. She sends the complete quantum circuit to Bob and allows him to measure all qubits in randomly chosen basis. Bob measures the received quantum circuit as a whole without knowing what bit sequence and basis selection Alice has chosen for sending the quantum information using quantum circuit in multiple shots. Let's take an example

of four qubits. When Bob receive quantum circuit of four qubits and measure it with randomly selected basis choice. There are $2^n = 2^4 = 16$ different possible states (bit strings) may appear as a result.

For the sake of simplicity, if Bob measures quantum circuit with single shot (by default in Qiskit number of shots are 1024) then only one state appears out of total sixteen possibilities. We store this state into a list for making a key, every character of this state is saved as a single and separate bit of information. During measurement, Bob saves all the basis choices into the list and same has already done on the other end i.e. Alice side (labeled as Alice's Basis and Bob's Basis). Finally, we compare the basis elements of both lists correspondingly and retain those bits as a final key for which basis are identical on both sides as shown in Table 1.

Alice's Key	1	0	1	0
Alice's Basis	X	X	Z	Z
Bob's Basis	X	Z	Z	Z
Bob's Key	1	0	1	0
Final Key	1		1	0

Table 1: Protocol of executing a final key when Alice shares her basis information with Bob on public channel.

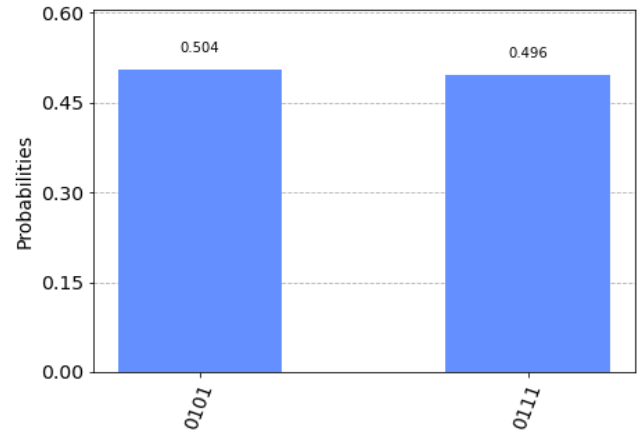


Figure 1 Measurement results of quantum circuit for four qubits on Bob end when no intruder is between Alice and Bob. (Execution on Classical Simulator)

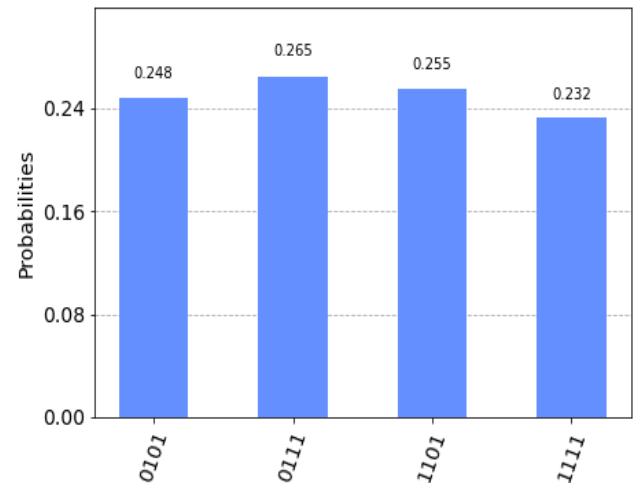


Figure 2 Measurement results of quantum circuit for four qubits on Bob end when intruder (Eve) is between Alice and Bob. (Execution on Classical Simulator)

Now, in second experiment the Eve is involved as well. Eve intercepts the quantum channel and tries to decode quantum information into classical bits. After the Eve intercepted the quantum channel, the Bob is not able to receive correct information due to quantum principles, as evident from the probability distribution of binary signals shown in Figure 2.

B. Experiments using IBM-Q

The IBM-Q provides the facility to run quantum algorithm/circuits on real quantum computer. We run our algorithm on IBM-Q machine and the Figure 3 shows the results of first experiment, when Alice send data to Bob without Eve in between. We can see in this case when we execute our quantum circuit on real quantum computers there are two types of peaks in histograms big and small peaks representing the high and less probabilities respectively. The big peaks are due to actual quantum circuit measurement on Bob's end whereas the small probabilities are result of noise effect in real quantum computer device due to quantum coherence effects. This noise effect shows that in case of simulation on actual quantum computer there are some probabilities getting wrong or misleading results.

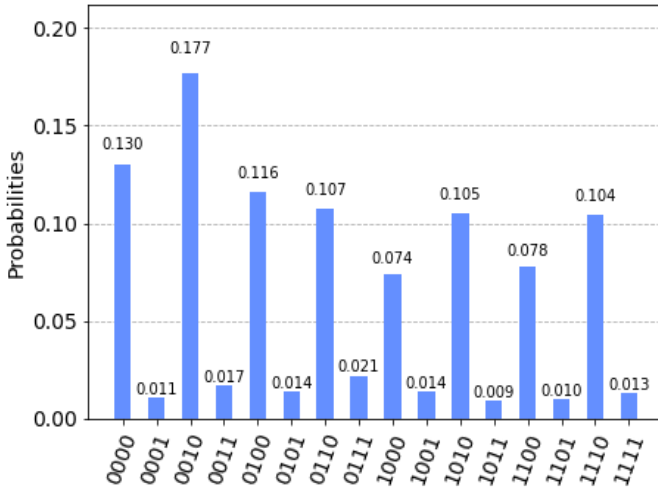


Figure 3 Measurement results of quantum circuit for four qubits on Bob end when no intruder in between Alice and Bob. (Execution on Quantum Computer, IBM Simulator).

When the Eve is introduced in between then the Bob results are modified, as shown in Figure 4, which proves that there was intruder in between the communication channel because probability distributions are modified. Some probabilities which are without Eve higher are becomes lower and vice versa. The change in probability distribution is evident for some information leakage because for specific state shared by Alice there is specific outcome. This thing can also be seen in terms of basis, we compare the basis elements of both lists (Alice and Bob's list) correspondingly and retain those bits as a final key for which basis are identical on both sides as shown in Table 1. If there is an intruder in between Alice and Bob then after comparing basis lists, the final accepted elements of list known as final key have an error and the final lists of both Alice and Bobs are not identical. Key exchange has been compromised so we will discard this key and repeat the process.

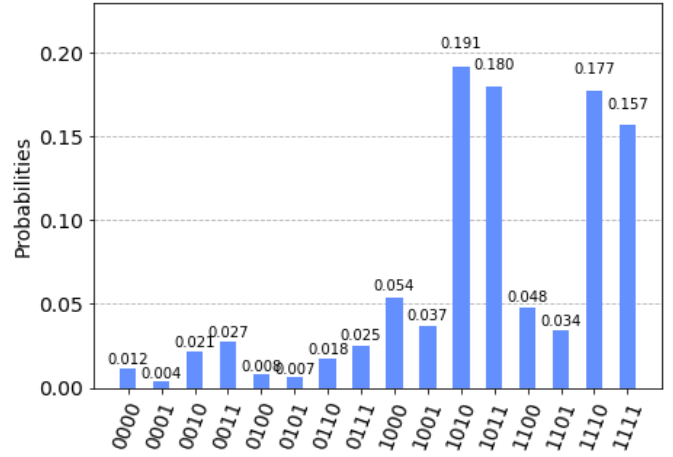


Figure 4 Measurement results of quantum circuit for four qubits on Bob end when intruder (Eve) in between Alice and Bob. (Execution on Quantum Computer, IBM Simulator).

VII. CONCLUSION AND FUTURE WORK

We have demonstrated proof of concept for simulation of BB84 protocol using Qiskit. We have performed experimentation on local as well as cloud simulators: (i) The "qasm_simulator" was used as local machine simulator, and (ii) the "ibmqx2" simulator was used as IBM cloud quantum simulator. The outcome of local as well as IBM simulator with and without an eavesdropper, between the two parties involved in key exchange were recorded and presented in the results section.

As there are a lot of limitations of the quantum computers right now so when, in future, quantum computers are advanced enough there are two main targets that can be achieved. First, the code should be changed a little and implemented on two quantum machines and an actual key is exchanged between two users over a real quantum computers network. Second, we are easily able to utilize only up to 4-qubits right now. So, in future, we should be able to utilize up to 512-qubits or 1024-qubits or even up to the needs of the users. It should be pretty easy with the provided code because the only thing that needs to change is the value of 'n'. But what can be done is once the key has been exchanged and a few bits are sacrificed to check the authenticity and integrity, some technique can be used to once again increase the key length. The code has to change or even few new lines can be written to implement this new logic.

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